

PATENT SPECIFICATION

Multi-Spectrum Anomaly Detection and Failure-Mode Correction System for Machine Learning Classifiers

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1. TECHNICAL FIELD

The present invention relates to unsupervised anomaly detection and post-hoc correction of deployed machine learning classifiers. More particularly, it provides a two-stage framework in which a Universal Deviation Layer (UDL) simultaneously measures deviations across multiple independent mathematical law domains to construct an attribution-rich anomaly tensor, and a Blind Spot Decomposition Theory (BSDT) layer uses that tensor to detect and correct structural failure modes in any deployed classification model without modifying the model's parameters. Applications include fraud detection in financial, aerospace, and medical domains.

2. BACKGROUND

2.1. Technical Problems

2.1.1. Limitation of Single-Perspective Anomaly Detection

Standard anomaly detectors (Isolation Forest, Local Outlier Factor, One-Class SVM, autoencoders) each produce a single scalar anomaly score. They identify *that* an observation is anomalous but not *which mathematical aspect* makes it anomalous. Because each detector imposes its own implicit representation of normality, an observation appearing normal under one representation may simultaneously violate a different mathematical law. No single detector spans the full space of mathematical anomaly, and none provides per-dimension attribution. Furthermore, achieving high AUC (area under the ROC curve) does not guarantee high coverage—anomaly scores may concentrate in a narrow band, failing to discriminate among subtypes (the “coverage–AUC gap”).

2.1.2. Structural Failure Modes in Deployed Classifiers

Supervised machine learning models deployed for fraud detection fail to detect certain fraudulent observations for structural reasons arising from specific, identifiable deficiencies in the model's representation of data space. Retraining addresses failures only if fresh

labelled data is available, which it typically is not (fewer than 0.1% of blockchain transactions are labelled as illicit). Ensemble methods do not diagnose *why* specific instances are missed and may share the same structural blind spot. Cost-sensitive reweighting is a global adjustment that does not distinguish between failure modes. Classical bias-variance decomposition operates in error space, not in feature space, providing no per-instance diagnostic information of why a specific observation was missed.

2.2. Prior Art Limitations

Known prior art includes:

- Isolation Forest (Liu et al., 2008); Local Outlier Factor (Breunig et al., 2000); One-Class SVM (Schölkopf et al., 1999): single-score general-purpose detectors providing no attribution.
- Deep autoencoders and out-of-distribution detection methods (ODIN, energy-based OOD): based on a single latent representation; no multi-law decomposition.
- SHAP and LIME: explain individual predictions in terms of input features but do not identify structural failure patterns or provide correction mechanisms.
- Conformal prediction: provides calibrated prediction sets but does not diagnose failure modes.

None provides simultaneous multi-law-domain anomaly tensor construction, MDN decomposition, boundary-centred dimension magnification, four-component failure mode decomposition, or post-hoc correction operators that operate on a frozen model.

3. SUMMARY OF THE INVENTION

The invention provides a unified two-stage framework. The shared inventive concept is *projecting observations through mathematically independent law domains to produce an attribution-rich deviation decomposition, and using that decomposition to discriminate and correct anomalies and classification failures without model retraining*. The primary technical contributions are:

- (i) The **Universal Deviation Principle**: a formal theorem that if a family of spectrum operators $\{\Phi_k\}$ is deviation-separating (the stacked Jacobian has full column rank), no anomalous observation can simultaneously match normal data across all operators. This provides a coverage guarantee absent from all single-perspective detectors.
- (ii) A **multi-spectrum anomaly tensor** $T_{i,k,d}$ constructed by projecting each observation through at least five independent mathematical law domains: statistical, dynamical (chaos), spectral (frequency), geometric, and exponential.
- (iii) A **Magnitude–Direction–Novelty (MDN) decomposition** separating how much (r_i), what type ($\hat{D}_i$), and how novel (θ_i) each deviation is.
- (iv) A **Boundary-Centred Dimension Magnifier** rescaling each tensor dimension by its empirical class-separability, amplifying well-separated dimensions and suppressing overlapping ones, using a single global hyperparameter set.

- (v) A **four-component failure mode decomposition** (BSDT) of the anomaly tensor into near-orthogonal components: Camouflage (C), Feature Gap (G), Activity Anomaly (A), and Temporal Novelty (T), serving as a per-instance diagnosis of why a deployed model fails.
- (vi) **Self-calibrating weight computation** producing a composite blind-spot score (MFLS) from the four components using mutual information, Fisher variance ratio, Bayesian online updating, or gradient optimisation.
- (vii) **Post-hoc correction operators** (additive, SignedLR, QuadSurf, ExpGate) that apply targeted adjustments to the deployed model’s predicted probabilities based on the failure profile, recovering missed detections without model retraining.
- (viii) A formal **orthogonality verification** of the four components (mean NMI < 0.065 , all VIF < 1.45) confirming that each captures an independent aspect of model failure.

4. DETAILED DESCRIPTION

4.1. Stage 1: Universal Deviation Layer (UDL)

4.1.1. Universal Deviation Principle

Definition (Deviation-Separating Family): An operator family $\{\Phi_k\}_{k=1}^K$ is *deviation-separating* if the stacked Jacobian:

$$J(x) = \begin{bmatrix} D\Phi_1(x) \\ \vdots \\ D\Phi_K(x) \end{bmatrix} \quad (1)$$

has full column rank for all x in the data domain.

Theorem 1 (Local Deviation Guarantee): If $\{\Phi_k\}$ is deviation-separating, then for any anomalous point x^* not identical to the normal centroid, there exists at least one operator under which x^* deviates from the projection of normal data.

Theorem 2 (Global Deviation Guarantee): On a compact anomaly set, a quantitative lower bound on minimum deviation across all operators can be established.

Theorem 3 (Finite-Sample Guarantee): Given n normal training samples the detection guarantee holds with probability $\geq 1 - \delta$ for a sample complexity depending on the operator family’s covering number.

Theorem 4 (Information-Optimal Fusion): Optimal weights for combining operator outputs are the solution to a semidefinite programme.

Theorem 5 (Greedy Operator Selection): A greedy algorithm selecting the most informative operator subset achieves a $(1 - 1/e)$ approximation to the optimal subset under a submodular information criterion.

4.1.2. Spectrum Operators

At least five mathematical law domains are instantiated as spectrum operators Φ_k :

Statistical Operator (Φ_{stat}). Measures distributional deviations: Shannon entropy of the local neighbourhood; Kullback–Leibler divergence from the reference normal distribution; Hellinger distance; skewness and kurtosis of local feature distributions.

Dynamical Operator (Φ_{chaos}). Measures temporal dynamics: Lyapunov exponent from the temporal context; recurrence rate from recurrence plot analysis; approximate entropy measuring regularity.

Spectral Operator (Φ_{freq}). Measures frequency-domain deviations: power spectral density divergence; frequency ratio (low-frequency to high-frequency energy); spectral entropy; spectral centroid shift.

Geometric Operator (Φ_{geom}). Measures geometric relationships: Mahalanobis distance from the reference centroid; cosine similarity to the reference direction; norm ratio.

Exponential Operator (Φ_{exp}). Amplifies small deviations invisible under linear representations:

$$\Phi_{\text{exp}}(x) = \exp\left(\alpha \cdot \frac{x - \mu}{\sigma}\right) \quad (2)$$

where μ, σ are reference statistics.

In an enhanced embodiment, up to 14 composable operators are available including phase-space, topological, kernel, rank, and wavelet operators. An operator correlation audit confirms that at most one pair exceeds the redundancy threshold ($\rho < 0.85$).

4.1.3. Anomaly Tensor Construction

Each observation x_i is projected through all K operators to yield a full anomaly tensor:

$$T_{i,k,d} \quad \text{for observation } i, \text{ operator } k, \text{ dimension } d \quad (3)$$

4.1.4. Magnitude–Direction–Novelty Decomposition

The tensor is decomposed into three interpretable components:

- (a) **Magnitude** $r_i = \|T_i\|$: total deviation amount.
- (b) **Direction** $\hat{D}_i = T_i / \|T_i\|$: unit tensor indicating deviation type.
- (c) **Novelty** θ_i : angular distance of $\hat{D}_i$ from reference deviation directions in training data.

4.1.5. Boundary-Centred Dimension Magnifier

Each tensor dimension is rescaled:

$$r'_d = \tanh\left(k_d \cdot \frac{r_d - b_d}{\sigma_d}\right) \quad (4)$$

where b_d is the decision boundary in dimension d , σ_d is the boundary-region scale, and $k_d = \gamma \cdot \delta_d$ is the steepness with δ_d computed from three complementary measures:

1. Cohen’s d (effect size) between normal and anomalous distributions;
2. class-distribution overlap area; and
3. per-dimension AUC.

The magnifier amplifies dimensions where classes are well-separated and suppresses dimensions where they overlap. In the preferred embodiment $\gamma=5.0$ and QDA regularisation $\lambda=10^{-4}$ constitute the single global hyperparameter set used across all datasets.

4.1.6. Classification Stage

Four classification strategies, with hybrid auto-selection:

- (a) **Fisher projection:** LDA projection followed by thresholding.
- (b) **QDA:** Quadratic discriminant analysis with regularised per-class covariance.
- (c) **QDA-Magnified:** QDA on the magnifier-rescaled tensor (preferred strategy).
- (d) **RankFusion:** Fully unsupervised rank-statistic fusion.

Hybrid auto-selection evaluates all strategies via cross-validation and selects the one maximising coverage while maintaining AUC above a floor.

4.1.7. Centroid Estimation

Seven centroid strategies (sample mean, trimmed mean, geometric median, MCD, medoid, Huber M-estimator, density-weighted centroid) are provided, with auto-selection minimising within-class scatter.

4.1.8. Coverage Metric

$$\text{Coverage@}k = \frac{|\{x : \text{rank}(x) \leq k, x \in \text{anomaly}\}|}{|\text{anomaly}|} \quad (5)$$

Coverage addresses the AUC–coverage gap. In the preferred embodiment mean coverage exceeds 91%.

4.2. Stage 2: Blind Spot Decomposition Theory (BSDT) Layer

The BSDT layer receives the anomaly tensor from Stage 1 (or may operate directly on classifier outputs) and decomposes each observation’s failure profile into four near-orthogonal components.

4.2.1. Relationship to Stage 1

The BSDT four components correspond to specialised instantiations of the UDL operator domains applied to a deployed classifier’s feature space. BSDT Camouflage (C) corresponds to the geometric operator applied to the legitimate-class centroid; Feature Gap (G) corresponds to the statistical operator measuring informational sparsity; Activity Anomaly (A) corresponds to the dynamical operator measuring deviation from known fraud activity profiles; Temporal Novelty (T) corresponds to the spectral and dynamical operators applied over the temporal axis of the training distribution.

4.2.2. Camouflage Component (C)

Measures proximity of the observation to the legitimate-class distribution:

$$C(x) = 1 - \frac{\|x - \mu_{\text{legit}}\|_2}{d_{\text{max}}} \quad (6)$$

where μ_{legit} is the legitimate-class centroid and d_{max} is the maximum centroid distance in the training data. $C(x) \approx 1$ indicates high camouflage: the observation mimics legitimate patterns, causing the model to under-estimate its suspicion.

4.2.3. Feature Gap Component (G)

Measures informational sparsity:

$$G(x) = \frac{|\{j : |x_j| < \varepsilon\}|}{d} \quad (7)$$

where d is the feature dimension and $\varepsilon = 10^{-8}$. High $G(x)$ indicates that many features are uninformative, depriving the model of discriminative signal.

4.2.4. Activity Anomaly Component (A)

Measures deviation of transactional activity from the activity profile of previously detected fraud:

$$A(x) = \sigma\left(\frac{\log(1 + |\text{activity}(x)|) - \mu_{\text{ref}}}{\sigma_{\text{ref}}}\right) \quad (8)$$

where $\text{activity}(x)$ is a transactional activity metric, μ_{ref} and σ_{ref} are statistics of the reference detected-anomaly population, and σ is the logistic sigmoid. This component captures fraud operating at activity scales the model has not learned to associate with fraud.

4.2.5. Temporal Novelty Component (T)

Measures the degree to which the observation is temporally novel relative to training data:

$$T(x) = \sigma\left(\frac{1}{2}(\bar{m}(x) - 2)\right) \quad (9)$$

where $\bar{m}(x)$ is the mean distance to the k nearest temporal neighbours in the training set, normalised by the temporal span.

4.2.6. Self-Calibrating Weight Computation

The four components are aggregated into a composite blind-spot score (MFLS):

$$\text{MFLS}(x) = \sum_{i \in \{C, G, A, T\}} w_i \cdot S_i(x) \quad (10)$$

The weights w_i (normalised to sum to 1) are computed automatically by one or more of:

- (a) Mutual Information between each component and the binary label on a held-out calibration set;

- (b) Fisher Variance Ratio (between-class to within-class variance);
- (c) Bayesian Online updating as labelled observations arrive; or
- (d) Gradient-based optimisation maximising detection performance.

4.2.7. Correction Operators

Four correction operators apply targeted adjustments to the frozen model's predicted probabilities $p(x)$:

Additive Correction.

$$p^*(x) = p(x) + \lambda \cdot \text{MFLS}(x) \cdot (1 - p(x)) \cdot \mathbb{I}[p(x) < \tau] \quad (11)$$

Logistic Regression (SignedLR).

$$p^*(x) = \text{clip}(p(x) + \beta_0 + \sum_i \beta_i S_i(x), 0, 1) \quad (12)$$

Quadratic Surface (QuadSurf).

$$p^*(x) = \text{clip}\left(p(x) + \phi_2([C, G, A, T])^\top \hat{\beta}, 0, 1\right) \quad (13)$$

where ϕ_2 is the degree-2 polynomial feature map (raw components, squared components, pairwise products) and $\hat{\beta}$ are coefficients fitted by ridge regression ($\alpha=1.0$).

Exponential Gate (ExpGate).

$$p^*(x) = \text{clip}(p(x) + g(x) \cdot q(x), 0, 1) \quad (14)$$

where $q(x)$ is the QuadSurf output and $g(x) = \sigma(v^\top [C, G, A, T] + b)$ is a learned sigmoid gate.

4.2.8. Seven Supplementary Features

In an enhanced embodiment, seven supplementary features supplement the core components for use by correction operators: the four raw components C, G, A, T ; the composite MFLS score; the predicted-vs-actual residual $|p(x) - y|$ (available in supervised calibration); and the interaction term $C(x) \cdot A(x)$.

4.2.9. Orthogonality Verification

The four components are verified near-orthogonal by three independent measures:

- Normalised Mutual Information: mean NMI $\in [0.051, 0.064]$ across all pairs, well below the 0.30 redundancy threshold.
- Variance Inflation Factor: all VIF < 1.45 , confirming negligible multicollinearity.
- PCA: all four components carry meaningful variance; none is dominated.

4.2.10. Cross-Domain Transfer

The adaptive damping function derived from the composite blind-spot energy:

$$\gamma^*(E) = \frac{E}{E + \theta} \quad (15)$$

is also applicable as a viscosity regularisation term in Navier–Stokes fluid dynamics simulations, representing a novel finance-to-physics transfer and linking the BSDT framework to the systemic risk controller described in related Application 1 (AMTTP/AF).

4.3. Cross-Domain Performance

Domain-dependent dominant components confirm genuine structural differences across application domains:

Domain	Dataset	Dominant Component
Blockchain	Elliptic	Temporal Novelty (T)
Blockchain	Ethereum Addresses	Temporal Novelty (T)
Banking	Credit Card Fraud	Activity Anomaly (A)
Banking	IEEE-CIS	Activity Anomaly (A)
Aerospace	Shuttle	Camouflage (C)
Medical	Mammography	Camouflage (C)
Pattern Recognition	Pendigits	Feature Gap (G)

4.4. Performance Characteristics

UDL — Single hyperparameter set ($\gamma=5.0$, $\lambda=10^{-4}$):

Dataset	QDA-Magnified AUC	Hybrid AUC
Synthetic	1.000	0.999
Mimic (camouflage adversarial)	1.000	0.999
Mammography	0.947	0.911
Shuttle	0.989	0.991
Pendigits	0.777	0.960
Mean	0.943	0.972

BSDT — Evaluated across 792 variant–model–dataset combinations:

Metric	Value
Best variant (QuadSurf) mean F1	0.787 (+43% relative)
False positive reduction	88.1% (44,740 $\rightarrow$ 5,345)
Maximum recall recovery	+84.4 percentage points
Shuttle rescue (F1)	0.134 $\rightarrow$ 0.968
Shuttle FP reduction	99.94%
Zero-missed-fraud configurations	10 / 12 model–dataset combos
Cross-domain credit card MFLS AUC	0.885
Cross-domain aerospace MFLS AUC	0.982
Cross-domain medical imaging MFLS AUC	0.916

5. CLAIMS

The following claims define the scope of protection sought. Where a claim refers to another claim, the dependency is by claim number. Features of any dependent claim may be combined with any independent claim or with other dependent claims.

1. A computer-implemented method for detecting and characterising anomalies in data, the method comprising:
 - (a) receiving an input observation comprising a feature vector;
 - (b) projecting the observation through a plurality of spectrum operators, each computing deviation measurements under a different mathematical law domain, the plurality comprising at least:
 - (i) a statistical operator measuring distributional deviations;
 - (ii) a dynamical operator measuring temporal dynamical deviations;
 - (iii) a spectral operator measuring frequency-domain deviations;
 - (iv) a geometric operator measuring geometric relationship deviations; and
 - (v) an exponential operator measuring deviations under exponential amplification;
 - (c) constructing a multi-dimensional anomaly tensor from the deviation measurements across all spectrum operators;
 - (d) decomposing the anomaly tensor into magnitude, direction, and novelty components, wherein the magnitude measures total deviation amount, the direction measures deviation type, and the novelty measures angular distance of the deviation direction from reference directions in the training data;
 - (e) applying a boundary-centred dimension magnifier that rescales each dimension of the anomaly tensor by a per-dimension steepness coefficient $k_d = \gamma \cdot \delta_d$ where δ_d is a discriminability score computed from empirical class-separability in that dimension; and
 - (f) classifying the rescaled anomaly tensor to produce an anomaly score and per-law-domain attribution identifying which mathematical law domains contribute to the anomaly determination.
2. A computer-implemented system for detecting anomalies and correcting failure modes in deployed machine learning classifiers, the system comprising:
 - (a) a plurality of spectrum operator modules collectively satisfying a deviation-separating condition such that the stacked Jacobian of the operators has full column rank, each module computing deviation measurements under a different mathematical law domain;
 - (b) a tensor construction module configured to assemble the deviation measurements into a multi-dimensional anomaly tensor;
 - (c) a decomposition module configured to decompose the anomaly tensor into magnitude, direction, and novelty components;

- (d) a magnifier module configured to rescale each dimension of the anomaly tensor using a boundary-centred rescaling function with per-dimension steepness computed from empirical class separability;
- (e) a classifier module configured to produce an anomaly score and per-law-domain attribution from the rescaled tensor;
- (f) a failure-mode decomposition module configured to compute, for each input observation and an associated predicted probability output by a deployed frozen classification model, four near-orthogonal failure mode components comprising:
 - (i) a camouflage component measuring proximity of the observation to a legitimate-class centroid;
 - (ii) a feature gap component measuring the proportion of near-zero features;
 - (iii) an activity anomaly component measuring deviation of a transactional activity metric from a reference detected-anomaly population; and
 - (iv) a temporal novelty component measuring temporal distance from training data patterns;
- (g) a weighting module configured to compute component weights using one or more of mutual information, Fisher variance ratio, Bayesian online updating, or gradient-based optimisation, and to aggregate the weighted components into a composite blind-spot score; and
- (h) a correction module configured to apply one or more correction operators to the predicted probabilities of the deployed model based on the computed component values, to produce corrected probabilities;

wherein the failure-mode decomposition module and correction module operate on a frozen deployed model without requiring modification or retraining of that model.

3. A computer-implemented method for detecting and correcting structural failure modes in a deployed frozen machine learning classification model without modifying the model, the method comprising:
 - (a) receiving a feature vector x and a predicted probability $p(x)$ output by the deployed model;
 - (b) computing a camouflage component $C(x)$ measuring proximity of the feature vector to a legitimate-class centroid in feature space;
 - (c) computing a feature gap component $G(x)$ measuring the proportion of features having values below a near-zero threshold;
 - (d) computing an activity anomaly component $A(x)$ measuring deviation of a transactional activity metric from the activity profile of a reference detected-anomaly population;
 - (e) computing a temporal novelty component $T(x)$ measuring the degree to which the observation is temporally distant from patterns present in the training data;
 - (f) computing a composite blind-spot score by aggregating the four components using automatically calibrated weights; and
 - (g) applying a correction operator to the predicted probability $p(x)$ based on values of one or more computed components to produce a corrected probability $p^*(x)$.

4. A method according to claim 1, wherein the boundary-centred dimension magnifier applies the rescaling function:

$$r'_d = \tanh\left(k_d \cdot \frac{r_d - b_d}{\sigma_d}\right)$$

where $k_d = \gamma \cdot \delta_d$, γ is a global steepness parameter, and δ_d is a per-dimension discriminability score computed from Cohen's d , class-distribution overlap, and per-dimension AUC.

5. A method according to claim 1, wherein classifying the rescaled tensor uses quadratic discriminant analysis with regularised per-class covariance matrices applied to the rescaled anomaly tensor.
6. A method according to claim 1, further comprising a hybrid auto-selection step that evaluates a plurality of classification strategies on cross-validation folds and selects the strategy maximising a coverage metric while maintaining AUC above a minimum threshold, wherein the coverage metric is:

$$\text{Coverage@}k = \frac{|\{x : \text{rank}(x) \leq k, x \in \text{anomaly}\}|}{|\text{anomaly}|}$$

7. A method or system according to any of claims 1 to 6, further comprising an operator correlation audit verifying independence of the spectrum operators by confirming pairwise correlation coefficients are below a redundancy threshold.
8. A method or system according to any of claims 1 to 7, further comprising a centroid auto-selection module selecting from a plurality of centroid estimation strategies the strategy minimising within-class scatter of normal data in the rescaled space.
9. A method or system according to any of claims 1 to 8, wherein a single global hyperparameter set is used across all datasets without per-dataset tuning.
10. A method or system according to any of claims 1 to 9, further comprising a greedy operator selection step that selects a subset of operators from a larger candidate set achieving a $(1 - 1/e)$ approximation to the optimal subset under a submodular information criterion.
11. A method according to claim 3, wherein the camouflage component is:

$$C(x) = 1 - \frac{\|x - \mu_{\text{legit}}\|_2}{d_{\text{max}}}$$

12. A method according to claim 3, wherein the feature gap component is:

$$G(x) = \frac{|\{j : |x_j| < \varepsilon\}|}{d}$$

13. A method according to claim 3, wherein the activity anomaly component is:

$$A(x) = \sigma \left(\frac{\log(1 + |\text{activity}(x)|) - \mu_{\text{ref}}}{\sigma_{\text{ref}}} \right)$$

14. A method according to claim 3, wherein the correction operator is a QuadSurf correction:

$$p^*(x) = \text{clip}\left(p(x) + \phi_2([C, G, A, T])^\top \hat{\beta}, 0, 1\right)$$

where ϕ_2 is the degree-2 polynomial feature map and $\hat{\beta}$ are ridge-regression coefficients.

15. A method according to claim 3, wherein the correction operator is an additive correction:

$$p^*(x) = p(x) + \lambda \cdot \text{MFLS}(x) \cdot (1 - p(x)) \cdot \mathbb{I}[p(x) < \tau]$$

16. A method according to claim 3, wherein the correction operator is an exponential gating correction comprising a QuadSurf output modulated by a learned sigmoid gate $g(x) = \sigma(v^\top [C, G, A, T] + b)$.
17. A method or system according to any of claims 3 to 16, wherein the four components are verified near-orthogonal by having normalised mutual information below 0.30 for all pairs and variance inflation factors below 2.0 for all components.
18. A method or system according to any of claims 3 to 17, wherein the dominant component varies by application domain: temporal novelty in blockchain domains, activity anomaly in banking domains, and camouflage in aerospace and medical domains.
19. A method or system according to any of claims 3 to 18, further comprising the step of computing seven supplementary features comprising the four raw components, the composite blind-spot score, a predicted-vs-actual residual, and a camouflage-activity interaction term.
20. A method or system according to any preceding claim, wherein the adaptive damping function $\gamma^*(E) = E/(E + \theta)$ derived from the composite blind-spot energy is applied as a viscosity regularisation term in a computational fluid dynamics simulation.

6. DRAWINGS

[To be filed separately: (1) Architecture diagram of the two-stage UDL + BSDT pipeline; (2) Diagram of the five spectrum operators and the anomaly tensor construction; (3) Visualisation of the MDN magnitude–direction–novelty decomposition; (4) Schematic of the boundary-centred dimension magnifier with per-dimension discriminability; (5) Operator correlation matrix confirming independence; (6) Four-component BSDT decomposition diagram; (7) Flow diagram of the correction operator pipeline; (8) NMI matrix and VIF chart confirming near-orthogonality; (9) Cross-domain performance comparison chart; (10) Schematic of BSDT operating on a frozen deployed classifier.]

ABSTRACT

A two-stage computer-implemented framework detects anomalies and corrects structural failure modes in deployed machine learning classifiers without model retraining. Stage 1 (Universal Deviation Layer) projects each observation through at least five independent mathematical law domains, constructs a multi-dimensional anomaly tensor, decomposes it into magnitude, direction and novelty components, and applies a boundary-centred dimension magnifier to produce an attributed anomaly score. A formal deviation guarantee proves that no anomaly can simultaneously match normal data across all law domains. Stage 2 (Blind Spot Decomposition Theory) decomposes each observation's failure profile into four near-orthogonal components: camouflage, feature gap, activity anomaly and temporal novelty. A composite blind-spot score with self-calibrating weights drives post-hoc correction operators that adjust a frozen model's predicted probabilities, recovering up to 84.4 percentage points of recall. The framework is validated across financial, aerospace and medical datasets using a single hyperparameter set.